LESSON PLAN

HIGH SPEED FLIGHT (12)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Various checks on propeller on the ground during ground run
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Various checks on propeller on the ground during ground run. General precautions to be observed while working on propeller, ground run
5. Specific Objectives	12.1.2.1 Trainee will be able to explain propellers and its associated maintenance practice

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Conduct ground run checks on propeller; observe safety precautions; instructor supervises procedure
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the general precautions for working on a propeller?
(b) Why is a ground run performed on a propeller?
(c) List the key checks required during a ground run.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• General precautions for propeller maintenance • Safety measures during ground run • Pre-run inspection requirements • Engine start and idle procedures • Propeller rotation observation	L & Ds	PPT, IB	15 min
(b)	• Blade tracking and balance checks • Vibration monitoring during run • Oil seal and hub inspection • Governor operation verification • Post-run shutdown procedures	L & Ds	PPT, PDF	20 min
(c)	• Recording ground run data • Defect identification criteria • Reporting maintenance discrepancies • Final clearance sign-off • Storage and protection measures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the critical safety precautions during a propeller ground run.
(b) Describe the procedure for identifying defects during the ground run.

12. Home Assignment and Follow Up	Prepare a brief note on the specific checks required for the next lesson plan (LP 02) regarding propeller in-flight performance.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min